



Major Article

Utilizing behavioral theories to explain hospital visitors' observed hand hygiene behavior

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Key Words:

Hand hygiene
Observed behavior
TPB
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TDF

Background: Hand hygiene is essential for infection prevention. This study aimed to find a suitable theoretical model and identify critical facilitators and barriers to explain hospital visitors' hand hygiene practice.

Methods: Visitors in 4 hospitals were observed and asked to give explanations for using or not using the hand rub dispenser. The written explanations of N=838 participants were coded according to three theoretical models: Theory of Planned Behavior, Health Action Process Approach (HAPA), and Theoretical Domains Framework (TDF).

Results: Self-reported hand hygiene behavior differed from observed behavior, with 15.75% wrongly claiming to have cleaned their hands. Critical facilitators for hand hygiene were *attitude toward the behavior, subjective norm, outcome expectancies, risk perception, planning, action control, knowledge and skills, motivation and goals, and social influences*. Key barriers included *perceived behavioral control; barriers and resources; memory, attention, and decision processes; and environmental context and resources*.

Conclusions: Visitors' self-reported hand hygiene behavior is over-reported. Both HAPA and TDF were identified as suitable theoretical models for explaining visitor's hand hygiene practice. Future behavior change interventions should focus on (1) visibility and accessibility of cleaning products; (2) informing laypeople about their role regarding infection prevention; and (3) leveraging social influence processes.

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BACKGROUND

Hand hygiene is a core element of infection prevention strategies. Cleaning hands with an alcohol-based hand rub is considered the most effective method to avoid cross-transmission of pathogens via touch.¹ Reducing the transmission of bacteria and viruses is especially important in health care facilities, which accommodate people who are most vulnerable to infections. Worldwide, hospitals struggle with health care-associated infections, which pose a serious threat to the health and safety of patients.^{2,3} The fact that hand hygiene is an important preventive measure against pathogen transmission and health care-associated infection is known by many health care

workers^{4,5} and laypeople.⁶ However, the hand hygiene practice of professionals⁷ as well as patients and visitors^{8,9} is generally insufficient. Countless interventions to enhance hand hygiene behavior in hospitals have shown that significant improvement is hard to achieve and even more difficult to sustain over time.¹⁰

One explanation for the ineffectiveness of many past interventions could be that their design was not grounded in theory.⁷ An increasing amount of evidence shows that theory-driven behavior change interventions are more effective in facilitating positive long-term effects.¹¹ Better understanding the facilitators and barriers of hand hygiene behavior in hospitals from a theory-driven perspective could help to design effective interventions. Some efforts in this direction have already been made, with several studies having identified theory-based predictors of hand hygiene behavior among health care workers.^{5,12,13,14} However, the facilitators and barriers of patients' and hospital visitors' hand hygiene behavior are not

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well-researched, despite studies showing that laypeople are a risk factor for the transmission of pathogens that can lead to infections,¹⁵ and improving their hand hygiene behavior reduces health care-associated infection rates.¹⁶ Therefore, the main goal of the present study is to investigate determinants for adequate hand hygiene behavior among laypeople in health care facilities.

However, studying predictors of human behavior is demanding, and several issues should be taken into consideration. First, the way in which a behavior is measured can have an influence on what kind of facilitators and barriers emerge as being important to predict it.¹⁴ Many previous studies relied on self-reported hand hygiene behavior rather than directly observed behavior (see ref. 7 for a systematic review). Using self-reports often is the only feasible way to gather data, as observing a large sample in hospitals is extremely labor-intensive and can be felt as obtrusive for the observed people. However, previous research has shown discrepancies between self-reported and observed hand hygiene behavior, with people generally overestimating how often they clean their hands.^{14,17} Therefore, the present study aimed to test whether laypeople's self-reported hand hygiene behavior in hospitals differs from their observed behavior.

Second, choosing a theoretical foundation to explain behavior is challenging. The literature is saturated with different behavioral theories, often with similar or overlapping constructs.¹⁸ Therefore, in the present study, we tried to identify a suitable theoretical model to explain laypeople's hand hygiene in hospitals from three promising candidates in the relevant literature: the Theory of Planned Behavior (TPB,¹⁹); the Health Action Process Approach (HAPA,²⁰); and the Theoretical Domains Framework (TDF,¹⁸). All 3 are well-validated theoretical models that have been used to study hand hygiene behavior before.^{6,12–14,21,22} Definitions of the theoretical models' constructs can be found in Table 1.

Third, the methodology used to measure determinants of hand hygiene behavior might influence which determinants will surface as the most important facilitators and barriers. Previous studies applying the same theoretical model identified slightly dissimilar predictors for hand hygiene behavior, especially when using different methods (qualitative vs quantitative) to measure the constructs.^{12,22} It is possible that respondents are not aware of factors influencing their behavior when providing explanations in an unstructured open-answer format. However, it is also possible that researchers fail to include important components in a structured questionnaire. Therefore, in the present study, we compared coded responses laypeople gave in an open-answer format to those from a previous study that used a structured questionnaire format, to see if facilitators and barriers of hand hygiene behavior that emerge in both answer formats are similar. These findings should help health care practitioners to design and evaluate future interventions to improve laypeople's hand hygiene behavior in hospitals.

METHODS

Design

A cross-sectional study was employed to investigate visitors' explanations for using or not using hand rub dispensers. The study was approved by the management of the involved hospitals. All participants gave oral consent to complete a questionnaire. The responsible Research Ethics Committee does not require a full ethical review of psychology studies with low risk.

Data collection

The study took place at three German hospitals, ranging in size from a small countryside clinic to a large university hospital. Convenience sampling was used to select one hospital from each of the 3

Table 1
Theoretical models and construct definitions

Model	Constructs*
Theory of Planned Behavior (TPB)	Attitude toward the behavior: Beliefs about the positive and negative outcomes of performing a behavior, and the evaluation of these outcomes. Subjective norm: Perceptions about normative expectations of significant others regarding a behavior and a person's motivation to comply with these. Perceived behavior control: Beliefs about the ease or difficulty involved in performing a behavior. Intention: An expression of readiness to perform a behavior.
Health Action Process Approach (HAPA)	Self-efficacy: Perceived capability to perform a behavior. (Divided into constructs of task self-efficacy, maintenance self-efficacy, and recovery self-efficacy). Outcome expectancies: Assessment of the benefits and disadvantages of performing a behavior. Risk perception: Perceived likelihood of experiencing a negative outcome in relation to a behavior, and perceived severity, which is the degree of harm arising from the negative outcome. Intention: Explicit decisions to act in a certain way. Planning: Specified details about the "when", "where", and "how" to act; and strategies on how to overcome anticipated barriers. Barriers and resources: eg, environmental constraints and social support. Action control: Comprises of self-regulatory effort, self-monitoring, and awareness of behavioral standards to adjust a behavior.
Theoretical Domains Framework (TDF) [†]	Knowledge: State of being familiar with or aware of something. Skills: Ability to do something which is developed through training and practice. Social/professional role and identity: Set of beliefs, attitudes, and characteristic behaviors expected of an individual in a social or work setting. Beliefs about capabilities (self-efficacy): Subjective perceptions of the capability to perform a behavior in a given setting or to attain desired results. Beliefs about consequences (anticipated outcomes): Subjective perceptions of the outcomes and consequences of a behavior in a given setting. Motivation and goals (intention): Willingness to exert effort in pursuit of a goal or outcome an individual wants to achieve. Memory, attention, and decision processes: Ability to retain information, focus on specific aspects of the environment, and choose between alternatives. Environmental context and resources: Conditions in a situation or an environment that facilitates or hinders the occurrence of a behavior. Social influences (norms): Interpersonal processes that facilitate a change in thoughts, feelings, or behavior. Emotions: A complex reaction pattern, by which the individual attempts to deal with a personally significant matter or event. Behavioral regulations: Processes aimed toward managing or changing a behavior.

NOTE. All definitions are based on original publication with regard to the theoretical models and definitions from the American Psychological Associations' Dictionary of Psychology.²³

*TDF constructs are called domains by the model's authors¹⁸ but will be referred to as constructs to ensure consistent use of language.

[†]The original TDF version¹⁸ without the construct "nature of behavior" was employed to make possible the comparison of the coded open-answer responses to those from a previous study that used a structured questionnaire.²¹ Definitions and results for the revised TDF version²⁴ can be found in the Supplementary Materials.

available care levels in the state (primary to tertiary care). Data were collected at the hospital's lobbies, where a team of trained research assistants observed whether people entering the building used the

hand rub dispensers. Previous studies have also used hospital lobbies to observe visitors' hand hygiene behavior,^{8,25} because visitors should sanitize their hands before patient contact, and the lobby's dispenser provides the first possibility for them to do so. The observers were located at a distance to the sole available dispenser in each lobby, allowing them to monitor the behavior unobtrusively. People who visually appeared to be visitors (eg, not attired in scrubs, hospital gowns, loungewear or pajamas) were approached several meters behind the dispenser by the same research assistant who observed their behavior; the research assistant then asked the person whether they came as a visitor. Already admitted patients were excluded as they have little reason to use the dispenser at the lobby. Newly arriving patients were also omitted because straining them with a questionnaire unrelated to their medical condition seemed inappropriate. Individuals who confirmed their status as visitors and to having not partaken in this study before were asked to participate. Visitors who consented to take part in the study received a questionnaire which, unbeknownst to them, was marked with a symbol indicating whether they had used the dispenser or not. Participants were not informed about the meaning of the symbol to avoid biasing their self-report. The visitors were asked to fill out the questionnaire on the spot in the lobby. Data were collected from December 2017 to November 2019.

Materials

The survey, estimated to take less than 15 minutes to complete, included information regarding the study's purpose and instructions. This was followed by questions about demographics, previous experiences with health care-associated infections, and whether the patient they visited had a heightened infection risk. Participants were asked if they had sanitized their hands at the lobby's sole hand rub dispenser right after entering the hospital and to explain their reasons for their behavior in an open-answer format (qualitative approach). Finally, participants were presented with questionnaire items to measure the constructs of the 3 theoretical models TPB, HAPA, and TDF (quantitative approach). The results of the quantitative approach are reported in a separate paper.²¹

Participants

Overall, 1,040 hospital visitors returned the questionnaire to the observers or left it at the reception. Of these, 838 (age ranged 18–91 years; Mean = 49.02, standard deviation = 16.77; 59.1% female) provided an explanation for using or not using the hand rub dispenser in the hospital lobby and were included for further analysis (a drop-out analysis can be found in the Supplementary Materials). Most participants (73.5%) had no previous experiences with health care-associated infections. When asked whether the patient they visited had a heightened infection risk, 53.7% said no, 28.3% affirmed the question, while 14.8% were not sure. A detailed sample description stratified by the hospitals in which the data was collected can be found in the Supplementary Materials (Table S2).

Data analysis

The goal was to identify facilitators and barriers of visitors' hand hygiene behavior from their immediate explanations for using or not using the hand rub dispenser. Therefore, the research team developed three sets of codebooks—one for each theoretical model—to classify responses into the constructs.¹ Each construct was coded as 1 = *mentioned* or 0 = *not mentioned* in the explanation. A similar

approach to a theory-guided coding of qualitative data has been used in previous studies.^{6,12} The open-answer explanations were coded according to these rulebooks independently by the same three trained raters. Every visitor's explanation for using or not using the hand rub dispenser was coded separately for each theoretical model. Coding every response 3 times allowed for drawing direct comparisons between the 3 theoretical models. The explanations were coded to more than one construct whenever it was deemed necessary (eg, when the response included more than one reason). The construct *intention* (relevant for TPB and HAPA) was not coded, as it was considered unlikely that a person would express an explicit intention to use the dispenser as a reason for doing so. A review of all explanations confirmed this assumption. The inter-rater reliability was calculated using the intraclass correlation coefficient, which is a commonly used index in inter-rater reliability analyses between 2 or more raters. All intraclass correlation coefficient values were between 0.79 and 1.00, indicating good to excellent inter-rater reliability. The remaining disagreements were discussed until a consensus was reached. The McNemar test, applicable for testing differences on a dichotomous dependent variable in a paired sample, was used to determine whether visitors' self-reported hand hygiene behavior (yes/no) differed from the observed behavior (yes/no). Binomial logistic regressions, typically deployed for regression analyses with dichotomous dependent variables, were used to test which constructs are relevant for predicting dispenser usage (yes/no). All statistical analyses were run in R version 3.6.3. The study's data and R-script will be made available online upon publication: <https://osf.io/5vq94/>

RESULTS

Hand hygiene behavior

According to the observation, 223 (26.6%) participants used the hand rub dispenser at the hospital, while 615 (73.4%) did not. When asked if they had sanitized their hands at the lobby's hand rub dispenser right after entering the hospital, 355 (42.4%) visitors affirmed that they had, while 483 (57.6%) said they had not. Accordingly, 15.8% of the visitors claimed to have cleaned their hands at the hand rub dispenser in the lobby, while the observation data show that they did not. Visitors' self-reported hand hygiene behavior in the hospital lobby differed significantly from the observed behavior ($\chi^2 = 130.01$, $P < .001$). Participants with a discrepancy between observed and self-reported behavior were removed from further analysis, leaving a total of $N = 706$ visitors.

We tested if age, gender, previous experiences with health care-associated infections, and the visited patient's infection risk could predict hand rub dispenser usage in the hospital lobby. All variables were included in a logistic regression with observed hand hygiene as the criterion. Only previous experience with health care-associated infections emerged as a significant predictor ($P < .001$). People who reported previous experience were more likely to clean their hands (Table 2).

Theoretical models predicting hand hygiene behavior

Table 3 shows the numbers of visitors' explanations coded to each theoretical model, along with examples. If an explanation did not fit to one of the theoretical models' constructs, it was coded as "other." Fewer explanations had to be coded as "other" for theoretical models that had more constructs (TPB: 3 constructs and 50.5% of explanations coded to "other"; HAPA: 6 constructs and 26.5% "other"; and TDF: 11 constructs and 16.5% "other"). This means that more detailed theoretical models allowed more explanations to be allocated to existing constructs.

¹ The 3 codebooks are in German language but can be made available in English upon request.

Table 2
Logistic regression results using observed hand hygiene behavior as the criterion

Predictor	b	SE	z	P	OR	95% CI for OR	
						LL	UL
Constant	−1.19	0.29	−4.05	<.001	0.31	0.17	0.54
Age	0.00	0.01	0.09	.930	1.00	0.99	1.01
Gender (female)	0.19	0.18	1.08	.280	1.21	0.86	1.72
Experience (yes)	0.67	0.19	3.56	<.001	1.95	1.35	2.81
Infection risk							
Heightened risk (yes)	0.22	0.19	1.15	.252	1.24	0.85	1.81
Unclear risk (don't know)	0.15	0.25	0.61	.540	1.17	0.71	1.89

NOTE. b, unstandardized coefficient; SE, standard error; z, b/SE, P, probability value; OR, odds ratio; LL, lower limit; UL, upper limit.

−2LL = 803.25, Model $\chi^2(5) = 16.28$, $P = .006$, AIC = 815.25, BIC = 842.17; $R^2 = 0.02$ (Hosmer-Lemeshow), 0.02 (Cox-Snell), 0.03 (Nagelkerke).

The constructs contained within the theoretical models can be considered as either facilitators or barriers for hand hygiene behavior. A construct is regarded as a facilitator when explanations coded into it were mainly given by dispenser users but not by nonusers. When the opposite is true, the construct is regarded as a barrier. Facilitators

that accounted for more than 10% of explanations were *attitude toward the behavior* (TPB), *outcome expectancies* (HAPA), *risk perception* (HAPA), *knowledge and skills* (TDF), *beliefs about consequences* (TDF), and *motivation and goals* (TDF). Barriers that accounted for more than 10% of explanations were *perceived behavioral control* (TPB); *barriers and resources* (HAPA); *memory, attention, and decision processes* (TDF); and *environmental context and resources* (TDF).

To test which constructs are relevant for predicting dispenser usage, 3 logistic regressions—one for each theoretical model—were calculated. Only constructs that were mentioned at least once by both dispenser users and nonusers were included as predictors. Table 4 shows the results of the 3 regressions. Including the constructs as predictors for observed hand hygiene behavior improved the fit of all 3 regression models compared to the baseline model with only the constant ($P < .001$). The Hosmer-Lemeshow goodness of fit tests, used to assess model calibration, showed no evidence for poor fit (all $P > .050$). Model discrimination assessed through area under the receiver operating characteristics (ROC) curve is excellent ($ROC_{TPB} = 0.87$, $ROC_{HAPA} = 0.92$, $ROC_{TDF} = 0.92$). The calculated model parameters suggest that the HAPA and the TDF both fit the data very well (see notes Table 4). Assessing the contributions of the individual predictors shows which constructs are most important for explaining

Table 3
Frequency (n and %) of written explanations for using/not using the hand rub dispenser coded to constructs of the 3 theoretical models with examples

Model	Construct	Dispenser users			Dispenser nonusers		
		n	%	Example	n	%	Example
Theory of Planned Behavior (TPB)	Attitude toward the behavior	158	20.8%	"I think it is right and I feel protected from bacteria [I]"	40	5.3%	"I don't like it; I'm afraid for my good micro-organisms"
	Subjective norm	17	2.2%	"It is our duty and out of consideration for others"	1	0.1%	"I just followed the flow of people"
	Perceived behavioral control	8	1.1%	"Placed directly at the entrance, easy to reach, [I]"	152	20.0%	"I could not find a hand rub dispenser"
	Other	62	8.2%	"Habit"	321	42.3%	"Totally forgot"
	Total	245	32.3%		514	67.7%	
Health Action Process Approach (HAPA)	Self-efficacy	2	0.2%	"[I] I can make a contribution to minimize infections"	0	0.0%	-
	Outcome expectancies	113	11.8%	"For my own safety and protection for others"	7	0.7%	"The skin burns from hand rub"
	Risk perception	129	13.5%	"Bacteria are everywhere, and I want to reduce the risk"	5	0.5%	"The danger only begins inside the hospital"
	Planning	32	3.3%	"Always use the dispenser when entering a hospital"	86	9.0%	"I always clean my hands in the room"
	Barriers and resources	8	0.8%	"It was recommended to me"	166	17.4%	"I did not see a dispenser; I did not see a sign"
	Action control	79	8.3%	"I know about the subject"	76	7.9%	"I didn't know about it"
	Other	42	4.4%	"It smells so good."	211	22.1%	"Don't know, [I]"
	Total	405	42.4%		551	57.6%	
	Knowledge and skills	154	14.62	"Prevention of nosocomial infections"	31	2.94	"I was not aware of the problem"
	Social/professional role and identity	15	1.42	"It should be taken for granted for all visitors [I]"	0	0.00	-
Theoretical Domains Framework (TDF)	Beliefs about capabilities	2	0.19	"[I] I can make a contribution to minimize infections"	4	0.38	"I had my hands full, so I could use the dispenser"
	Beliefs about consequences	116	11.02	"it [I] kills off bacteria"	25	2.37	"The skin burns from hand rub"
	Motivation and goals	106	10.07	"I want to protect myself from bacteria and viruses"	3	0.28	"Not in the mood"
	Memory, attention and decision processes	1	0.09	"Didn't really think about it [I]"	163	15.48	"Totally forgot"
	Environmental context and resources	25	2.37	"Placed directly at the entrance, easy to reach"	212	20.13	"I did not see a dispenser; I did not see a sign"
	Social influences	5	0.47	"It is our duty and out of consideration for others"	3	0.28	"I just followed the flow of people"
	Emotions	1	0.09	"Fear of infections"	4	0.38	"I don't like it; I'm afraid for my good micro-organisms"
	Behavioral regulations	2	0.19	"Mandatory"	7	0.66	"It's not specifically pointed out"
	Other	55	5.22	"Habit"	119	11.30	"I'll do it at the ward"
	Total	482	45.77		571	54.23	

NOTE. Knowledge and skills were combined into 1 construct.

Table 4
Logistic regression results using observed hand hygiene behavior as the criterion

Model	Predictor	b	SE	z	P	OR	95% CI for OR	
							LL	UL
Theory of Planned Behavior (TPB)*	Constant	−1.79	0.15	−11.73	<.001	0.17	0.12	0.22
	Attitude toward the behavior	3.18	0.23	13.66	<.001	24.08	15.41	38.45
	Subjective norm	4.44	1.07	4.14	<.001	84.99	15.47	1603.07
	Perceived behavioral control	−1.60	0.42	−3.82	<.001	0.20	0.08	0.43
Health Action Process Approach (HAPA) [†]	Constant	−2.57	0.24	−10.74	<.001	0.08	0.05	0.12
	Outcome expectancies	2.86	0.51	5.57	<.001	17.45	6.61	50.59
	Risk perception	4.38	0.52	8.35	<.001	79.70	30.89	249.88
	Planning	0.78	0.30	2.61	.009	2.18	1.21	3.91
	Barriers and resources	−1.19	0.47	−2.54	.011	0.31	0.11	0.72
	Action control	2.13	0.28	7.76	<.001	8.45	4.98	14.68
Theoretical Domains Framework (TDF) [‡]	Constant	−0.72	0.18	−4.12	<.001	0.49	0.34	0.68
	Knowledge and skills	1.87	0.36	5.16	<.001	6.47	3.25	13.53
	Beliefs about capabilities	−0.64	1.11	−0.58	.564	0.53	0.03	3.62
	Beliefs about consequences	−0.74	0.43	−1.73	.083	0.48	0.20	1.08
	Motivation and goals	4.04	0.84	4.81	<.001	57.04	13.79	423.43
	Memory, attention and decision processes	−4.74	1.09	−4.34	<.001	0.01	0.00	0.05
	Environmental context and resources	−1.63	0.29	−5.67	<.001	0.20	0.11	0.34
	Social influences	2.11	1.08	1.97	.049	8.28	1.08	86.45
	Emotions	−1.84	1.19	−1.55	.122	0.16	0.01	1.27
	Behavioral regulations	−1.55	1.16	−1.33	.184	0.21	0.02	1.86

NOTE. b, unstandardized coefficient; SE, standard error; z, b/SE, P, probability value; OR, odds ratio; LL, lower limit; UL, upper limit.

*−2LL = 525.39, Model $\chi^2(3) = 355.29$, $P < .001$, AIC = 533.39, BIC = 551.63; $R^2 = 0.40$ (Hosmer-Lemeshow), 0.40 (Cox-Snell), 0.55 (Nagelkerke).

[†]−2LL = 419.63, Model $\chi^2(5) = 461.06$, $P < .001$, AIC = 431.63, BIC = 458.98; $R^2 = 0.52$ (Hosmer-Lemeshow), 0.48 (Cox-Snell), 0.67 (Nagelkerke).

[‡]−2LL = 428.76, Model $\chi^2(9) = 451.92$, $P < .001$, AIC = 448.76, BIC = 494.36; $R^2 = 0.51$ (Hosmer-Lemeshow), 0.47 (Cox-Snell), 0.66 (Nagelkerke).

the behavior. All TPB and HAPA constructs included in the regressions made a significant contribution to the prediction of observed hand hygiene behavior. In the TDF regression model, the constructs *knowledge and skills*; *motivation and goals*; *memory, attention, and decision processes*; *environmental context and resources*; and *social influences* emerged as significant predictors.

DISCUSSION

The data show that hospital visitors' self-reported hand hygiene behavior differed significantly from their observed behavior. In total, 15.8% of visitors claimed to have sanitized their hands in the lobby while the observation showed that they did not. This finding is consistent with previous research showing that self-reported hand hygiene is usually overreported.^{14,17} Hand hygiene is morally loaded, which makes it prone to socially desirable responding. Therefore, relying on self-reporting to measure visitors' hand hygiene behavior is not ideal. Still, the majority of visitors reported their behavior correctly. Therefore, if cheap, unobtrusive, and widely applicable methods of accurately measuring hand hygiene behavior are not available, relying on self-reported data may be an acceptable alternative for research in this area.

All 3 theoretical models (TPB, HAPA, and TDF) proved useful for examining visitors' hand hygiene behavior. Visitors who cleaned their hands provided slightly more detailed explanations for their actions. Dispenser users predominantly gave explanations highlighting their awareness of being a risk factor for the transmission of dangerous pathogens and their effort to avoid negative consequences. Nonusers mainly blamed the hospital environment or the situational context for not being able to clean their hands. Most explanations from nonusers indicated that they knew that hand hygiene was appropriate but searched for reasons why they did not use the dispenser. People often ascribe reasons for their behavior in a manner that helps fulfill their desire for a positive self-image. This is known as *self-serving attributional bias*, which is the tendency to attribute positive events to ourselves and our own actions while attributing negative events to others and situational factors.²⁶ It is possible that many visitors were subject to this attribution bias. Very few

explanations were coded to the following constructs: *subjective norm*, *self-efficacy*, *social/professional role and identity*, *beliefs about capabilities*, *social influence*, *emotions*, or *behavioral regulations*. The respondents seemed to be unaware of or downplay the influence of social processes and emotions, which is consistent with other research and might reflect people's desire to provide rational explanations for their decisions.²⁷ *Perceived capability/self-efficacy* to perform hand hygiene was a largely irrelevant construct for the visitors, most likely because using a hand rub dispenser is such a simple task.

Considering the suitability of the three logistic regressions modeled according to the 3 theoretical models to predict dispenser usage, both HAPA and TDF fit the data well and better than the TPB. It is noteworthy that the HAPA's model fit was on par with TDF for 2 reasons. First, HAPA is more parsimonious in using constructs to explain behavior than TDF. Second, in previous work drawing from the same sample that also utilized a questionnaire for measuring the constructs to predict self-reported hand hygiene behavior,²¹ HAPA had a worse model fit and predicted less variance than TDF. The fact that TDF emerged as a suitable model is consistent with these previous results.²¹ Additionally, TDF allowed the most explanations to be allocated to existing constructs, showing some merit of a more comprehensive framework approach for understanding behavior. Having coinciding evidence from both a qualitative (ie, coded open-answer format) and a quantitative (ie, structured questionnaire) approach, TDF appears to be a solid theoretical model to explain and predict the facilitators and barriers of laypeople's hand hygiene practice in hospitals. However, it should be noted that TDF is theoretical framework and not a theory. Unlike TPB and HAPA, it does not propose any testable causal processes linking the theoretical constructs. Nevertheless, it has been extensively used for the implementation of evidence-based practice.

We now turn to discuss which components are most relevant to predict visitor's hand hygiene behavior for the 3 theoretical models separately:

- TPB: All 3 TPB constructs significantly predicted observed dispenser usage among visitors. Visitors giving explanations coded into the constructs *attitude toward the behavior* and *subjective norms* were more likely to use the hand rub dispenser. In contrast,

participants who provided explanations coded into the construct *perceived behavioral control* were less likely to use the dispenser. This is consistent with the findings of a previous study using a quantitative approach (structured questionnaire format), where all 3 constructs significantly correlated with either *intention* to clean hands or with self-reported hand hygiene behavior.²¹

- HAPA: The 5 HAPA constructs included in the regression model also significantly predicted observed hand hygiene behavior. Hospital visitors who gave explanations coded into the constructs of *outcome expectancies*, *risk perception*, *planning*, and *action control* were more likely to perform hand hygiene in the lobbies. Participants who gave explanations coded into the construct *barriers and resources* were less likely to use the hand rub dispenser. Since no explanation from nonusers was coded into the construct of *self-efficacy*, it was not included in the logistic regression. This made comparing the qualitative and quantitative HAPA results harder. Nevertheless, in both analyses, *outcome expectancies*, *risk perception*, and *action control* surfaced as important facilitators for hand hygiene behavior or *intention* as a mediator.²¹
- TDF: Not all TDF constructs emerged as significant predictors for visitors' dispenser usage. Observed hand hygiene was positively associated with visitors' explanations fitting into the constructs *knowledge and skills*, *motivation and goals*, and *social influences*. Visitors who wrote explanations related to *memory*, *attention*, and *decision processes* and *environmental context and resources* were less likely to use the dispenser. As with HAPA, a direct comparison of results from the qualitative and the quantitative approach is not possible, since *social/professional role and identity* was not included in the logistic regression. Still, *knowledge and skills*, *social influences*, and *memory*, *attention*, and *decision processes* emerged as important constructs for visitors' hand hygiene behavior in both analyses.²¹ These findings are also consistent with another recent study on patient's hand hygiene behavior.⁶

Considering these results, 3 implications for future behavior change interventions can be deduced: (1) Hand rub dispensers need to be placed at highly visible and easily accessible locations to prevent forgetting and facilitate their usage. (2) Interventions should be designed to raise awareness that laypeople are a risk factor for the transmission of dangerous pathogens and explain that adequate hand hygiene can reduce this risk. (3) Interventions should utilize social influence processes. For instance, informational materials and reminders could contain normative messages highlighting that hand hygiene is the norm.

From a methodological perspective, results from using coded open-answer explanations (qualitative approach) and structured questionnaire scales (quantitative approach) as predictors for visitors' hand hygiene behavior are not identical but reasonably consistent. This consistency indicates that both approaches seem to be reliable methods to investigate facilitators and barriers of laypeople's hand hygiene behavior.

Limitations

This study had some limitations. First, the original TDF-constructs have overlying components. For instance, the constructs *social/professional role and identity* and *social influences* both contain the components of *social norms* as well as *identity*. This makes a distinct coding of visitors' explanations into the constructs challenging. In cases with overlapping components, we coded the explanations to both constructs, which increases the risk for multicollinearity. However, we found no evidence for critical multicollinearity between the predictors (all VIFs between 1.01 and 1.71). Second, the explanations for using or not using the dispenser tended to be brief, not very detailed, and sometimes slightly ambiguous, which made the coding process

more challenging. However, the inter-rater reliability was very good, indicating that the raters coded the explanations consistently. Third, the study was conducted prior to the COVID-19 pandemic. It might be assumed that, due to the pandemic, hospital visitors are currently more aware of the importance of hand hygiene. More research is needed to validate the relevance of the identified facilitators and barriers after the crisis. Fourth, the visitors' explanations used for coding the predictors were given after the behavior occurred and might be a post hoc rationalization for their action. This is very hard to avoid even when predictors are measured before the occurrence of the outcome, since people often consider their past behavior when making judgments and decisions about a future action. Finally, several nonusers claimed to clean their hands later on at the ward or the patient room. Since the researchers did not follow visitors through the hospital, it cannot be determined if they actually did so. Future research should be conducted to observe visitors' hand hygiene behavior throughout the hospital.

CONCLUSIONS

The aim of the present study was to expand the understanding of hospital visitors' hand hygiene behavior. The analysis showed that visitors overreported their hand rub dispenser usage in comparison to direct observation, which provides further evidence that relying solely on self-reported data is insufficient when studying hand hygiene behavior. This was the first study to use visitors' open-response explanations coded according to 3 theoretical models as predictors for their observed dispenser usage. While all theoretical models helped to explain visitors' behavior, HAPA and TDF emerged as more useful than TPB. The identified facilitators and barriers of hand hygiene behavior lead to three important implications for future interventions: (1) placing dispensers in highly visible and easily accessible locations; (2) informing laypeople about their role in the infection prevention strategy; and (3) leveraging social influence processes. These implications should help practitioners design and evaluate interventions to improve visitors' hand hygiene behavior. Better hand hygiene practice can prevent health care-associated infections and improve safety for everyone in health care facilities.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at <https://doi.org/10.1016/j.ajic.2021.01.003>.

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